A Fast Multikernel Graph Convolutional Network for Hyperspectral Image Classification

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***Abstract*—Deep learning-based hyperspectral image (HSI) classification has significant applications in remote sensing scene understanding. Whole-image propagation classification methods can achieve high-precision classification at the full-pixel level with a significant efficiency advantage. However, such methods generally face challenges such as the smoothness of high-frequency details and difficulty in efficiently modeling deep features. To address these issues, this article proposes a fast multikernel graph convolutional network (FMGCN) for fast whole-image classification. The model designs a multikernel graph convolution (MKGC) algorithm to enhance the expression ability of deep spectral–spatial features. This algorithm generalizes the graph convolution process from a single kernel propagation process to a propagation process with an arbitrary number of kernels to more fully exploit spectral–spatial features in non-Euclidean spaces. In addition, a symmetric compressed refinement network (SCRN) is constructed to restore the weakened local structural information in the whole-image propagation, and an embedded edge compensation strategy (EECS) is combined to further strengthen the high-frequency features in the boundary regions. Based on this design, FMGCN is able to capture more comprehensive global dependency information in a single whole-image propagation, achieving a uniform balance of high accuracy and high efficiency. Experiments on four representative hyperspectral datasets show that FMGCN achieves the best classification performance, effectively alleviates over-smoothing in complex structural scenarios, and outperforms batch propagation models in terms of inference efficiency.**

***Index Terms*—Edge compensation, fast inference, fast multikernel graph convolutional network (FMGCN), hyperspectral image (HSI), whole-image propagation.**

I. Introduction

IN EARTH observation tasks based on remote sensing techniques, the captured data lack morphological details of ground-based objects. Especially when using RGB images for object detection, segmentation, change detection, and other scenes, it is difficult to accurately distinguish objects of the same color. With the continuous development of hyperspectral imaging [1], [2], remote sensing hyperspectral images (HSIs) can record tens or even hundreds of spectral bands in each pixel, providing a rich basis for discriminating ground-based objects with similar colors but different materials. As a result, HSI classification techniques are widely used in ecological monitoring [3], smart agriculture [4], mineral exploration [5], [6], defense security [5], and other sectors [7]. However, the high spectral dimensionality and spectral variability of HSIs often lead to feature redundancy, noise sensitivity, and increased computational burden, posing challenges to robust classification, especially in large-scale remote sensing scenes.

Early works are mostly based on hand-crafted spectral features and shallow classifiers such as support vector machines (SVMs) [8] and random forest models [9], [10], which are susceptible to noise and interclass overlap in high-dimensional feature spaces. With the development of deep learning, HSI classification methods based on convolutional neural networks (CNNs) [3], [11], [12] have been extensively studied. These methods take pixel-neighborhood patches as input, use 1-D/2-D/3-D CNNs to extract spectral–spatial joint features, and combine residual learning [12], [13], [14] or attention mechanisms [15], [16], [17] to improve representation power. However, most convolutional methods still rely on sliding windows or patch-level inference, which requires dense sliding windows on HSI when reasoning over large scenes, leading to significant computational overhead. At the same time, the limited receptive field and local modeling also easily lead to the lack of semantic consistency (salt-and-pepper noise) of the whole-image prediction.

To explicitly exploit the spatial topological priors contained in HSIs, graph convolutional networks (GCNs) [18], [19], [20] have been introduced into HSI classification. Related approaches typically construct graph structures based on pixels [21] or superpixels [22], [23], and perform feature propagation over non-Euclidean neighborhoods via graph convolution, effectively enhancing the modeling capabilities of intraclass consistency and long-range dependencies. Introducing a preliminary denoising or feature refinement module before or after the graph construction can enhance the representational ability of the graph structure. For example, shallow convolutional layers are applied to suppress spectral noise in the original HSI, while conditional random fields are further employed as a postprocessing step to refine pixel-level predictions and correct potential errors introduced by superpixel segmentation [24]. Some works further introduce attention

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graphs [25], hypergraphs [26], [27], [28], or multivariate correlation graphs [29] to characterize richer spectral–spatial relationships. However, existing graph models still suffer from two shortcomings. First, most graph filters use a single kernel, which makes it difficult to take into account the diverse topological patterns of different classes and regions under complex terrain distributions and multiscale structures. Second, the convolution of multirelational graphs is often accompanied by a large amount of computation, which introduces a significant computational and storage burden when directly applied to high-resolution whole graphs. As a result, most of the work is still stuck on patch-level or local-region modeling, and it is difficult to balance efficiency and overall graph consistency.

Recently, transformer-based models have received a lot of attention in natural image and HSI analysis [30], [31], [32]. Vision transformer (ViT) [33], [34] and its improved architectures model long-range dependencies at the global scale via a multihead self-attention mechanism; meanwhile, hierarchical Swin transformer [35] and hybrid CNN-transformer structures [11] are also used to balance local details and global context. However, transformer-based methods typically rely on patch blocking or window partitioning, which still suffer from blocking artifacts and discontinuous window boundaries. Moreover, multihead self-attention on high-resolution whole graphs is computationally expensive and sensitive to hardware resources, which remains a major limitation for fast classification applications of large-scale scene HSI. In parallel, state-space model-based architectures such as Mamba [36], [37], [38] have also been introduced into HSI classification to improve long-range dependency modeling with linear computational complexity. By replacing self-attention with selective state-space modeling, Mamba-based methods enable efficient global spectral–spatial interactions. It is worth noting that, similar to other whole-HSI modeling methods, Mamba also suffers from the risk of over-smoothing, making it difficult to accurately distinguish boundary features.

On the other hand, regardless of whether a convolutional backbone, a graph-based model, or a transformer architecture is employed, the structural sensitivity to ground-truth object boundaries, small patches, and elongated targets tends to degrade under the cumulative effect of repeated downsampling and feature compression. This phenomenon is particularly evident in HSI classification methods that rely on whole-image propagation, where multilevel compression and aggregation of features and adjacency representations are commonly introduced to control computational complexity [39]. To alleviate this issue, several existing approaches attempt to enhance structure awareness through edge-guidance modules [40], attention-based gating mechanisms [41], or postprocessing strategies [42]. Nonetheless, most of these methods either operate locally and remain weakly coupled to the overall compression-decompression pipeline or are applied as independent preprocessing steps. Consequently, they fail to provide continuous and efficient compensation for the boundary information loss within an end-to-end training framework.

In summary, existing methods for HSI classification still suffer from obvious deficiencies in multirelational spectral–spatial structure modeling, graph-level fast inference, and boundary information preservation. In addition, the HSI classification task also faces challenges such as high-dimensional spectra, limited labeled samples, and spectral variability. CNN-based methods generally rely on patch-level inference [43], which is difficult to balance between efficiency and graph-wide consistency. Although some global graph processing methods have been proposed, the over-smoothing of node features often occurs when aggregating high-order domain features [44]. While transformer-based methods have a strong ability to extract long-term dependencies, they also face high inference time and memory costs [45]. In particular, when multiple graph convolution operations are used to enhance structural representations, ensuring that the operations are lightweight while suppressing the loss of boundary information due to compression is one of the important challenges.

To address the aforementioned challenges, this article proposes a fast multikernel graph convolutional network (FMGCN) for HSI classification. Unlike traditional patch-level approaches, FMGCN directly takes the dimensionally reduced HSI as input and performs whole-image spectral–spatial feature modeling and classification in an end-to-end manner. The proposed framework employs a symmetric-reconstruction architecture to enable efficient whole-image propagation while preserving structural consistency. In this design, the classification accuracy, computational efficiency, and structural fidelity are jointly optimized by integrating multikernel graph convolution (MKGC) and embedded edge compensation strategy (EECS). The main contributions of this work are summarized below.

1) In the proposed FMGCN, we introduce an MKGC network (MKGCN) to capture deep regional relationship features. In this module, an MKGC algorithm is designed to overcome the limited representation capability of single-kernel graph filters in HSI classification. Unlike existing graph-based methods that rely on a single convolution kernel, MKGC performs parallel spectral–spatial feature propagation via multiple graph kernels on a compressed graph representation. MKGC is used to learn multiple spectral–spatial feature representations in parallel on compressed graphs, which enables the modeling of diverse topological patterns under complex land cover distributions, while significantly reducing the computational cost of whole-image graph convolutions.
2) An EECS is proposed to alleviate the loss of boundary information while ensuring lightweight operation. Aiming at the problem of edge and high-frequency information weakening caused by MKGC's multilevel compression of features, FMGCN extracts multiscale edge features in the preprocessing stage, and feeds them into the network in the compression and decoding stages in a structure-guided manner, so as to ensure the efficient operation of MKGC while supplementing high-frequency structural cues.
3) A symmetric compressed refinement network (SCRN) is constructed to achieve quadratic edge enhancement. FMGCN employs a symmetric-reconstruction architecture to implement forward inference at the whole-HSI

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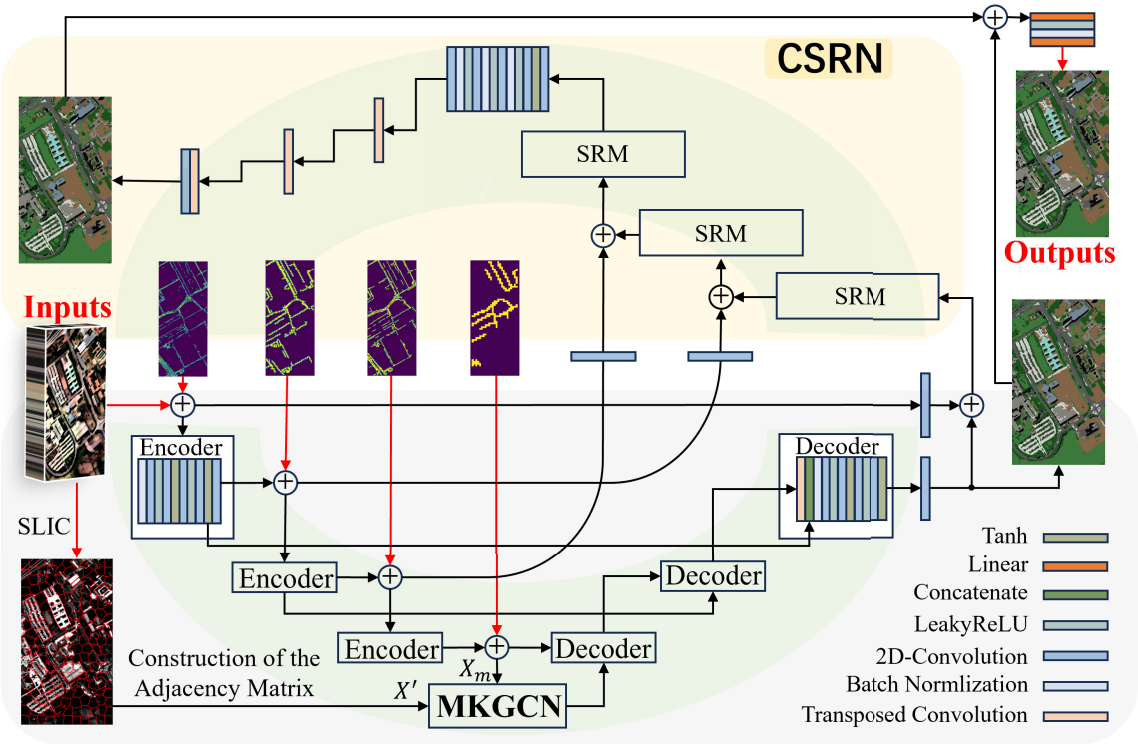


Fig. 1. Framework of FMGCN. The proposed model consists of an MKGCN, an EECS, and an SCRN architecture. The red arrows indicate the information flow for input/output. The black arrows indicate the network data information flow.

level, which significantly improves classification efficiency and global semantic consistency compared to patch-by-patch inference. After completing the entire HSI classification in the first stage, a spatial rearrangement module (SRM) is further introduced. This refinement stage compensates for boundary degradation caused by contextual aggregation during compression, resulting in improved edge quality and regional consistency in the final classification maps.

The remainder of this article is organized as follows. Section II details the overall framework of FMGCN, including MKGCN, EECS, and SCRN architectures. Section III presents the experiments and the resulting analysis. Section IV concludes the work of this article and discusses possible future research directions.

II. Proposed Method

The proposed network is presented in detail in this section. Fig. 1 shows the overall framework of the proposed FMGCN. This includes an MKGCN, an EECS, and an SCRN architecture. The inputs to the network are the entire HSI after PCA and normalization, the edge structure maps at multiple scales, and the adjacency matrix constructed from the superpixel segmentation results. Superpixel segmentation is performed using a simple linear iterative clustering (SLIC) algorithm. We feed the whole image into FMGCN in a single pass and design an MKGC method to more fully extract multiple types of spectral–spatial information. To reduce the high computational cost caused by multiple graph convolution operations, we employ convolution-superpixel cooperation to compress the feature maps and adjacency matrices. Since the downsampling operation inevitably leads to the loss of detailed information, we exploit the edge structure graph and compressed completion network architecture to compensate for the loss of high-frequency information as much as possible and thus achieve fast and high-precision classification of the entire HSI.

A. MKGC Network

In traditional GCNs, each layer's convolution operation is usually performed by a single learnable weight matrix for feature extraction. This approach is difficult to fully exploit the spectral information. Currently, some HSI classification models use multiple channels to establish the relationship between spatial and spectral graphs [29], [46], improving the limitations of single embedding. However, this is typically achieved by manually extracting features to construct a fixed number of adjacency matrices, which makes it difficult to freely learn and explore potential high-dimensional features. The MKGC algorithm aims to model various spectral–spatial information more fully and flexibly with multiple learnable convolution kernels in a CNN-like manner. Therefore, this algorithm can autonomously select the number, scale, and other information of the kernels and adaptively update the topology of the graph.

The computational complexity of the graph convolution operation is as follows:

$$\mathcal{O}\left(N_G F_{\text{in}} F_{\text{out}} + E_G F_{\text{out}}\right) \tag{1}$$

where N_G is the number of nodes, E_G is the number of edges, F_{in} is the input feature dimension, and F_{out} is the

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Algorithm 1 Superpixel-Based Graph Sample Construction

Require: Standardized HSI cube $\mathbf{X}_0 \in \mathbb{R}^{H\times W\times b}$ (after PCA reduction and normalization);
number of superpixels N_{sp}; compactness parameter λ.
Ensure: Superpixel-level feature matrix $\mathbf{F}_G \in \mathbb{R}^{N_{\text{sp}}\times b}$;
adjacency matrix $\mathbf{A} \in \mathbb{R}^{b\times N_{\text{sp}}\times N_{\text{sp}}}$.

// Step 1: Superpixel segmentation
$\mathbf{S} = \text{SLIC}(\mathbf{X}_0, N_{\text{sp}}, \lambda)$▷$\mathbf{S} \in \{1,\ldots,N_{\text{sp}}\}^{H\times W}$ is the superpixel label map
// Step 2: Superpixel feature aggregation
for $s = 1$ **to** N_{sp} **do**
  $\Omega_s = \{(i,j) \mid \mathbf{S}(i,j) = s\}$▷ pixel set of superpixel s
  $\mathbf{F}_G(s,:) = \frac{1}{|\Omega_s|}\sum_{(i,j)\in\Omega_s} \mathbf{X}_0(i,j,:)$
end for
// Step 3: Adjacency matrix generation on superpixels
Initialize $\mathbf{A} \in \mathbb{R}^{b\times N_{\text{sp}}\times N_{\text{sp}}}$
for $i = 1$ **to** N_{sp} **do**
  for $j = 1$ **to** N_{sp} **do**
    $\mathbf{A}(:,i,j) = \mathbf{F}_G(i,:) - \mathbf{F}_G(j,:)$
  end for
end for
return A

output feature dimension. This formulation states that the size of the adjacency matrix, which is the number of nodes, directly determines the computational complexity. Currently, GCN-based HSI classification models can be categorized into patch-level predictions and single-pass whole-image predictions. In patch-level predictions, samples can be used to construct adjacency matrices with a small number of nodes and to classify pixels pixel-by-pixel in a batch fashion, which reduces the computational cost. However, batch processing requires multiple data transfers and network operations during prediction, making it difficult to achieve fast whole-image classification. Single-pass whole-image prediction can obtain classification results for the entire image after one network propagation, which is more convenient and faster in practical applications. Correspondingly, this approach requires the construction of adjacency matrices for the entire HSI image, and large matrix multiplications increase the computational cost. Similar to multichannel CNNs, MKGC operates as an independent operation and does not rely on specific feature scales. In this article, single-pass whole-image prediction is used to balance the computational cost with the actual prediction time, and a series of compression strategies are used to reduce the computational load.

Equation (1) shows that the computational load is affected by the number of nodes and the dimensionality of the node embedding vector. Therefore, this article reduces the computational cost in both directions, namely, using superpixel segmentation to reduce the number of nodes and principal component analysis to reduce the dimensionality of the embedding vector. To make better use of the information of each component, the mean image of the segmented images with b principal components is obtained to construct the corresponding b-channel adjacency matrix $\mathbf{A}$. At the same time, considering the lightweighting of the algorithm and the local common spectral interference (clouds, temperature, etc.), each channel of the cluster adjacency matrix is constructed by the image of each principal component through difference. Let H and W denote the spatial dimensions of the input HSI cube, and b denotes the number of principal components. The adjacency matrix is constructed as follows.

The resulting adjacency matrix implements the region relation information at a relatively low computational cost. However, this tensor, as an adjacency matrix group, still has the following two problems. First, the node feature tensor used to obtain the adjacency matrix undergoes rigid information compression (SLIC), leading to irreversible information loss. Second, the current adjacency matrix is extracted manually and lacks learnability, making it impossible to update the topological structure. Therefore, we improve the input for graph convolution based on this work. Specifically, since $\mathbf{A}$ can already represent the relational information of the entire HSI, we perform convolutional compression on the HSI that has undergone PCA and normalization. In each convolutional compression unit, in addition to convolution, the main operation is to halve the edge length by a pooling operation. This approach can extract features while retaining the corresponding spatial information of the original data. As shown in Fig. 1, after three convolutional units, a simple scale alignment (edge cropping and padding) is performed so that the output feature map $\mathbf{X}'$ has the same size as the mean image. This feature map not only mitigates the rigid information loss, but also retains some degree of the original node relation information through the learnability of convolution and the spatial correspondence of pooling. It is worth noting that as an alternative node feature, $\mathbf{X}'$ still differs from $\mathbf{X}_m$. Therefore, the adjacency matrix must also be learnable. Moreover, the adjacency matrix group $\mathbf{A}$ encompasses b types of spectral–spatial relations (number of spectral bands). This restricts the convolutional compressed feature maps to also have the same number of channels. Both of these are independent processes, and there is no physical correspondence between each band, posing a potential risk of information conflict in this hard matching. Finally, while the constructed adjacency matrix group contains information from different bands, essentially only one spectral–spatial relation can be constructed for each band. The combination of multiple bands, while improving information utilization, still does not allow for a more complete and comprehensive parallel exploration of the rich spectral–spatial features of HSI. To eliminate the above issues and limitations, as shown in Fig. 2, we design an MKGC layer and build an MKGCN based on it. This module employs a multikernel feature extraction method and accordingly improves the propagation process of adjacency matrix groups, node feature tensors, multikernel weights, and graph convolutions.

As shown in the following equation, in this article, we first expand the dimension of $\mathbf{A}$, and replicate its k copies in the new dimension, which are used to interact with the subsequent k graph convolution kernels. Subsequently, it is fed into a 2-D convolutional layer such that the second dimension aligns the number of bands of $\mathbf{X}'$ (b channels) in an updateable manner

$$\mathbf{A}^{\text{rep}} \in \mathbb{R}^{K\times b\times N_{\text{sp}}\times N_{\text{sp}}}, \quad \mathbf{A}^{\text{rep}}_{k,:,:} = \mathbf{A} \quad \forall k \tag{2}$$

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$$\bar{\mathbf{A}} = \mathrm{Conv}_{\mathrm{adj}}\left(\mathbf{A}^{\mathrm{rep}}\right) \in \mathbb{R}^{K\times C_{\mathrm{in}}\times N_{\mathrm{sp}}\times N_{\mathrm{sp}}} \tag{3}$$

where $\mathbf{A}^{\mathrm{rep}}_{k,:,:}$ denotes the kth kernel's replicated adjacency matrix group, K represents the number of output channels/the number of kernels/the number of copies, and $\bar{\mathbf{A}}$ represents the adjacency matrix after the channel transformation, where C_{in} indicates the number of input channels. Equation (3) implements a learnable update of the multivariate correlation graph through convolutional operations. This enables learnable updates of the structural dependencies and supports an MKGC process that can capture relations across different receptive fields. To align with the previously introduced "multichannel adjacency tensor," we construct a replicated version of the node feature maps as follows:

$$\bar{\mathbf{X}_G} \in \mathbb{R}^{K\times C_{\mathrm{in}}\times N_{\mathrm{sp}}\times F_{\mathrm{in}}}, \quad \bar{\mathbf{X}_G}_{k,:,:,:} = \mathbf{X}_G \quad \forall k \tag{4}$$

$$\mathbf{X}_G = \mathrm{DT}\left(\mathbf{X}_m\right) \tag{5}$$

where $\bar{\mathbf{X}_G}$ represents the node feature corresponding to $\bar{\mathbf{A}}$, which is replicated from $\mathbf{X}_G$. $\mathbf{X}_m$ is one of the inputs of MKGCN in Fig. 1. DT$(\cdot)$ denotes the dimension transformation, which transforms the channel dimension of the feature map $\mathbf{X}_m$ into the embedding vector dimension, thereby constructing the node feature $\mathbf{X}_G$ of $w_X \times h_X$ nodes. w_X and h_X represent the length and width of the feature map $\mathbf{X}_m$, respectively. This replication operation initializes a multikernel node feature $\bar{\mathbf{X}_G}$, where each kernel processes node features under its own receptive field. This provides the prerequisite for multikernel adjacency convolution, in which the learnable kernel weights and adjacency maps jointly support more diverse information transmission. For a given kernel k, F_{out} represents the output feature dimension, and F_{in} corresponds to the input feature dimension. The multikernel spectral–spatial aggregation process is computed as

$$\mathbf{S}_{k,c,n,f} = \sum_{i=1}^{F_{\mathrm{in}}} \bar{\mathbf{X}_G}_{k,c,n,i}\, \mathbf{W}_{k,c,i,f} \tag{6}$$

where $\mathbf{S} \in \mathbb{R}^{K\times C_{\mathrm{in}}\times N_{\mathrm{sp}}\times F_{\mathrm{out}}}$ stores the kernel transformed features. For each kernel k and each feature channel c, we apply an independent linear projection $\mathbf{W}_{k,c,:,:}$ that maps the input feature dimension F_{in} to the output feature dimension F_{out} ($\bar{\mathbf{X}_G} \to \mathbf{S}$). This forms a set of multikernel, multichannel spectral–spatial embeddings. After adjacency compression and multikernel feature interaction, we obtain the aggregated graph feature $\mathbf{Y} \in \mathbb{R}^{K\times N_{\mathrm{sp}}\times F_{\mathrm{out}}}$. The output of the kth kernel can be written as

$$\mathbf{Y}_{k,n,f} = \sum_{c=1}^{C_{\mathrm{in}}} \sum_{m=1}^{N_{\mathrm{sp}}} \bar{\mathbf{A}}_{k,c,n,m}\, \mathbf{S}_{k,c,m,f} + \mathbf{b}_f \tag{7}$$

where $\mathbf{b}_f$ is the bias vector, and $\mathbf{Y} = \{Y^{(k)}\}_{k=1}^{K} \in \mathbb{R}^{K\times N\times F_{\mathrm{out}}}$. The above process demonstrates three essential aspects as follows.

1) For each kernel (k), the graph propagation simultaneously performs weighted aggregation over the band channels of the adjacency matrix (c) and the neighborhood of the node (m).
2) The adjacency weights $\bar{\mathbf{A}}_{k,c,n,m}$, derived from convolutional compression of multiband adjacency features, are functions of structure smoothing and relation update.
3) The support feature $\mathbf{S}_{k,c,n,f}$, which comes from the multikernel feature transformation, is a spectral–spatial embedding tailored to different kernels and channels.

As shown in Figs. 1 and 2, the input HSI undergoes three 4× downsampling operations to encode features and corresponding edge scale features, which are then fed into the MKGCN. The MKGCN contains three MKGC operations to extract deep non-Euclidean spectral–spatial relational features. After the MKGC is completed, stepwise decoding is performed to recover the original size. Each upsampling layer consists of a single transposed convolution and multiple convolutional blocks. The recovered feature images are fed into a 2-D convolution to output preliminary predictions for the entire HSI. During training, a part of the loss function is constructed using the pixel indices corresponding to the pixels in the training set and the predicted results, thus implementing the output of the entire image classification result in a U-Net-like form.

B. Embedded Edge Compensation Strategy

After the downsampling, MKGCN, and reconstruction processes, spatial smoothing occurs, resulting in the loss of fine boundaries and small object structures. To alleviate this issue, FMGCN incorporates an EECS. As illustrated in Fig. 1, the input HSI is first used to extract multiscale structural features, which are then injected into the compressed feature propagation pathway to compensate for the losses incurred by the compressed path.

1) Multiscale Edge Extraction: Given a standardized HSI cube $\mathbf{X}_0 \in \mathbb{R}^{H\times W\times b}$, we first perform multiscale edge detection to obtain a set of spatial-resolution-dependent edge maps

$$\mathbf{E}^{(s)} = \mathrm{EdgeDet}\left(\mathrm{PCA}\left(\mathbf{X}_0\right), s\right), \quad s \in \mathcal{P} = \{1, 4, 16, 64\} \tag{8}$$

where EdgeDet$(\cdot)$ denotes a Canny-based edge detector applied at scale s, and s denotes the downsampling factor with respect to the original resolution. The detector is configured with Gaussian smoothing, subsampling, and Canny-based gradient thresholding.

The resulting multiscale edge maps $\mathbf{E}^{(s)} \in \mathbb{R}^{H_s\times W_s}$ capture the intensity variations and structural boundaries at scale s, where H_s and W_s are spatial sizes consistent with the s-fold compression ratio (e.g., $s = 1$ corresponds to the original resolution, while $s = 4^n$ corresponds to the nth downsampling stage). As illustrated in Fig. 1, the extracted $\mathbf{E}^{(1)}$, $\mathbf{E}^{(4)}$, $\mathbf{E}^{(16)}$, and $\mathbf{E}^{(64)}$ correspond to the boundary information at successive downsampling stages and are used to provide boundary priors for each layer of the network during the embedding process.

2) Input-Level Edge Embedding: At the network input, we inject fine-grained boundary cues by embedding multiscale edge information into the original spatial domain before applying downsampling or compression on the features. The input-level embedded tensor is defined as

$$\tilde{\mathbf{F}}^{(0)}(c, i, j) = \mathrm{PER}\left(\mathbf{X}_0(i, j, c)\right) + \mathbf{1}\,\mathbf{E}^{(1)}(i, j) \tag{9}$$

where $\mathbf{1} \in \mathbb{R}^b$ is an all-one vector used to broadcast the edge intensity across all PCA channels. PER$(\cdot)$ denotes the transformation of the channel dimension to the first dimension.

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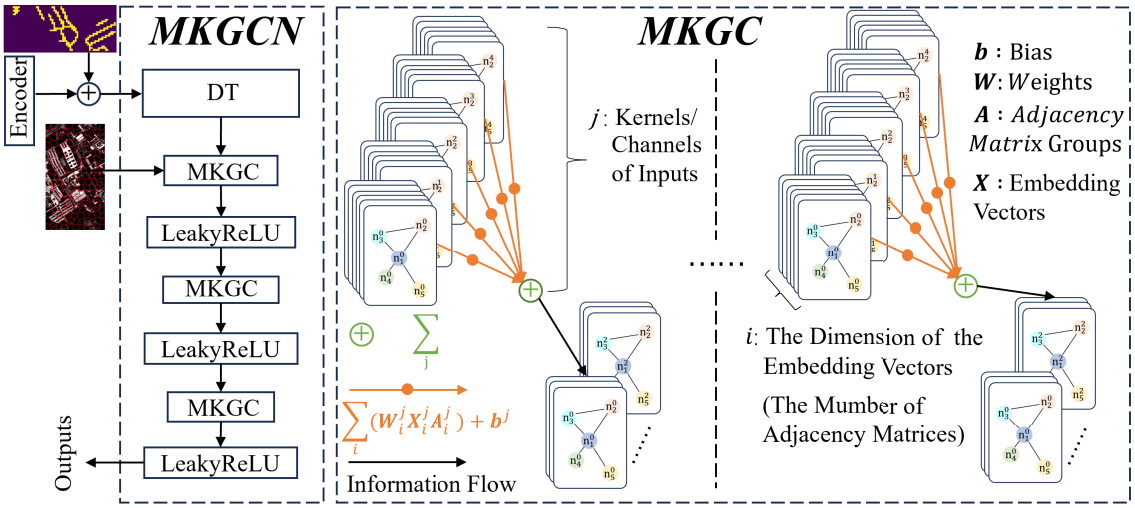


Fig. 2. Framework of MKGCN.

The resulting $\tilde{\mathbf{F}}^{(0)}$ is then fed into the first downsampling stage. This step treats multiscale edge cues as auxiliary high-frequency signals injected into the spatial domain of the image, enabling subsequent layers to preserve sharper boundary awareness during feature compression.

3) Multiscale Edge Embedding: As the feature representation is progressively compressed through multiple downsampling stages, the spatial resolution becomes lower while the receptive field expands. Such a process inevitably smoothens the spatial boundaries. To avoid the loss of structural information along the encoding process, FMGCN embeds spatial-resolution-matched edge maps into the feature tensors at each encoding stage, enabling a coordinated "compression-edge preservation-edge compensation" mechanism. Let the feature maps of the first, second, and third encoding stages be defined as $\mathbf{F}^{(1)} \in \mathbb{R}^{C_1 \times H_1 \times W_1}$, $\mathbf{F}^{(2)} \in \mathbb{R}^{C_2 \times H_2 \times W_2}$, and $\mathbf{F}^{(3)} \in \mathbb{R}^{C_3 \times H_3 \times W_3}$, where H_i and W_i denote the spatial dimension after the ith downsampling stage. C_i is the number of channels of the ith downsampling feature map. Correspondingly, we construct spatially aligned edge maps $\mathbf{E}^{(4^i)} \in \mathbb{R}^{H_i \times W_i}$. Using a simple size refinement operator $\mathcal{A}(\cdot,\cdot)$ to align (e.g., cropping and padding), the embedded edge-compensated features are written as

$$\tilde{\mathbf{F}}^{(1)}(c,i,j) = \mathbf{F}^{(1)}(c,i,j) + \gamma_4 \mathcal{A}\left(\mathbf{1}\mathbf{E}^{(4)}, \mathbf{F}^{(1)}\right)_{i,j} \tag{10}$$

$$\tilde{\mathbf{F}}^{(2)}(c,i,j) = \mathbf{F}^{(2)}(c,i,j) + \gamma_{16} \mathcal{A}\left(\mathbf{1}\mathbf{E}^{(16)}, \mathbf{F}^{(2)}\right)_{i,j} \tag{11}$$

$$\tilde{\mathbf{F}}^{(3)}(c,i,j) = \mathbf{F}^{(3)}(c,i,j) + \gamma_{64} \mathcal{A}\left(\mathbf{1}\mathbf{E}^{(64)}, \mathbf{F}^{(3)}\right)_{i,j} \tag{12}$$

where $\gamma_4, \gamma_{16}, \gamma_{64}$ are weighting coefficients corresponding to different spatial scales. Equations (10)–(12) embody the central idea of the EECS: at each downsampling level, scale-matched boundary cues are injected to preserve structural information. By maintaining such modulation throughout the encoding path, the framework can effectively compensate for boundary preservation effects even under strong compression.

It is worth noting that $\mathbf{F}^{(3)}$ is also used as input to the MKGCN ($\mathbf{X}_m$) and subsequent decoder. Thus, multiscale boundary cues not only affect convolutional features but also further propagate across the multikernel graph domain, enhancing structural sensitivity at both local and nonlocal levels.

C. Symmetric Compressed Refinement Network

Based on the MKGCN and EECS, FMGCN introduces an SCRN to perform global prediction refinement and structural detail correction. Specifically, FMGCN employs a two-branch design: 1) the first branch produces an initial prediction with EECS and MKGCN; and 2) the second branch embeds the boundary-aware encoded features and initial predictions into an SCRN, enabling progressive structure alignment and boundary preservation across multiple SRM and outputs the second prediction. The predictions of the two branches are summed to obtain the final classification result.

As illustrated in Fig. 1, the second branch consists of three levels of SRM, each including a downsampling operation and a corresponding feature update module. Refinement inputs include: 1) multiscale edge features extracted at the first branch and 2) the initial prediction map (or the outputs of the previous SRM). Each SRM processes its inputs through spatial rearrangement, convolution, and other operations to achieve downsampling while minimizing information loss.

1) Inputs Construction for the First Refinement Level: Let $\mathbf{R}_1 \in \mathbb{R}^{c_r \times H \times W}$ denote the initial prediction generated by the first branch, where c_r denotes the number of classes. The inputs of the first SRM ($\mathbf{X}_s^1$) incorporate both the initial prediction and the boundary-enhanced features

$$\mathbf{X}_s^1 = f_{\text{up}}(\mathbf{R}_1) + f_r^0\left(\tilde{\mathbf{F}}^{(0)}\right) \tag{13}$$

where $f_{\text{up}}(\cdot)$ and $f_r^0(\cdot)$ denote convolutional projection functions applied to align the spatial and channel dimensions.

2) Inputs Construction on the iTh SRM: Let $\mathbf{X}_s^{i-1}$ denote the rearranged feature tensor from the previous SRM, and $\mathbf{X}_s^i$ denote the inputs of the current SRM. The input construction on the ith SRM is defined as

$$\mathbf{X}_s^i = \mathbf{X}_s^{i-1} + f_r^{i-1}\left(\tilde{\mathbf{F}}^{(i-1)}\right) \tag{14}$$

where $\mathbf{X}_s^i$ represents the ith level downsampled feature map, and $f_r^i(\cdot)$ extracts structural boundary information

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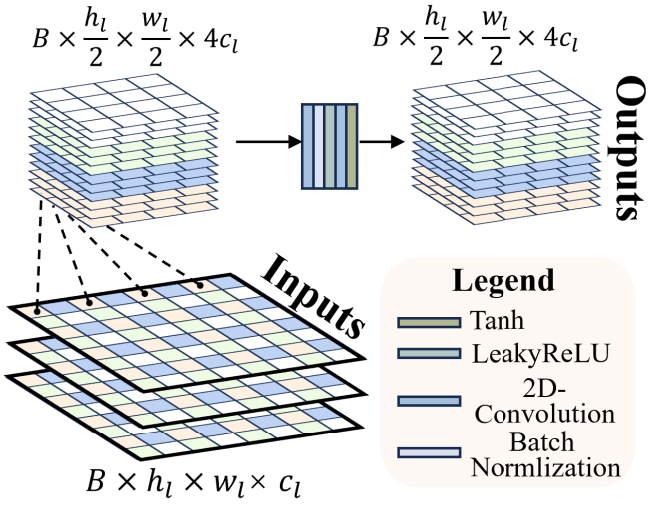


Fig. 3. Framework of SRM.

corresponding to the ith resolution level. The refinement process ensures that the SRM feature maps maintain consistency between the boundary information and the compressed representation at each level.

3) Spatial Rearrangement: The core of the SCRN lies in the spatial rearrangement operation. This operation resembles a reverse patch-merging process: it reduces the spatial resolution by deterministic reshaping while preserving the feature organization. As shown in Fig. 3, spatial reordering is similar to the patch merging operation. Notably, when combined with multiscale edge features, this rearrangement operation preserves high-frequency information more fully and thus compensates more accurately for the boundary sensing capability.

Through a cascade of symmetric rearrangement, convolution, nonlinear activation, and interlevel fusion, the network progressively extracts object boundaries and structural details. During this process, edge features act as guiding signals to modulate the refinement behavior, compensating for the structural information lost during the compression phase.

To restore the original resolution symmetrically in the spatial domain, the network achieves upsampling by cascading three transposed convolutions and one convolutional layer. Three $4\times$ upsampling operations based on transposed convolutions gradually recover the feature scale. Finally, a convolutional layer is used to fine-tune the size, ensuring that the feature size is strictly aligned with the original image, which is output as the prediction result of the SCRN branch $\mathbf{R}_2$.

Since the rearrangement operation is purely deterministic and does not introduce information loss, the decoder preserves the invertible character of the spatial transformation. This allows the network to maintain geometric consistency while still benefiting from nonlinear filtering. Consequently, the refined prediction $\mathbf{R}_2$ for the second branch captures information about the boundary structure that is complementary to that of the first branch. The outputs of the two branches are fused through a lightweight fusion module

$$\mathbf{R}_y = \mathcal{F}_{\text{late}}(\mathbf{R}_1 + \mathbf{R}_2) \tag{15}$$

where $\mathcal{F}_{\text{late}}$ denotes the fusion operator, which is implemented through a series of operations such as linear layers, activation functions, and batch normalization. This module adaptively reweights the channels of $\mathbf{R}_1$ and $\mathbf{R}_2$, producing a fused prediction $\mathbf{R}_y$ that aggregates complementary cues from both branches. During training, the loss is applied directly to $\mathbf{R}_y$ at the pixel level. During inference, a simple $\arg\max(\cdot)$ operation on $\mathbf{R}_y$ yields the final classification results.

Overall, the SCRN in FMGCN establishes a structurally constrained dual-branch framework. The synergy of these branches enables FMGCN to effectively compensate for edge losses while maintaining high-precision classification performance.

D. FMGCN

As illustrated in Fig. 1, FMGCN consists of three major components: an MKGCN, an EECS, and an SCRN. The input to the FMGCN is a PCA-standardized HSI together with a set of multiscale edge maps extracted by a Canny-based detector. The overall propagation pipeline of FMGCN is summarized as follows. First, the standardized HSI and multiscale edge information are jointly processed and forwarded into multiple spatial downsampling encoding units. Subsequently, several MKGC operations are performed, and the original spatial dimensions are recovered by multiple decoders employing transposed convolution (while also producing first-branch prediction). Next, these predicted features are aggregated with the outputs from each encoding stage and passed to the SCRN to obtain the second-branch prediction results. Finally, the predictions from both branches are fused through a lightweight fusion module to generate the final classification map. FMGCN performs a single forward propagation over the entire HSI, achieving $64\times$ downsampling coverage with dual-branch learning. These properties enable FMGCN to deliver faster and better classification performance based on GCN-style structure inference.

III. Experiments

In this section, we compare the proposed model with several representative models, including classification performance, the impact of key parameters, and ablation experiments. In addition, this section presents the relevant datasets, metrics, experimental setup, and the analysis of the results.

A. Datasets

We evaluate the proposed method on four benchmark HSI datasets that differ in spatial resolution, spectral properties, and scene complexity. The detailed category information and false color images of the datasets are shown in Table I.

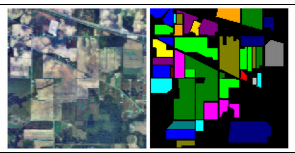
1) The Indian Pines (IPs) dataset was acquired over agricultural areas in northwestern Indiana, containing 145×145 pixels and 16 classes. The wavelength range spans 400–2500 nm, with a spectral resolution of 10 nm and a spatial resolution of 20 m. After removing water absorption bands, 200 spectral channels remain. The dataset is widely used for fine-grained crop classification tasks.

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2) The Salinas Valley (SV) dataset primarily covers agricultural fields. The image consists of 512 × 217 pixels and 16 labeled categories. After standard preprocessing, 204 effective spectral bands remain. The spatial resolution is 3.7 m, making the dataset suitable for precision agriculture research.
3) The Houston (HS) dataset was released during the 2013 IEEE Geoscience and Remote Sensing Society Image Analysis and Data Fusion Contest [47]. It captures a complex urban scene containing buildings, roads, parking lots, and vegetation. The image size is 349 × 1059 pixels, with 15 labeled categories. The spatial resolution is 2.5 m, and the spectral range spans 0.38–1.05 μm. This dataset is commonly used for large-scale urban and cross-domain HSI classification studies.
4) The Pavia University (PU) dataset was collected on the PU and contains a mixture of man-made structures and vegetation. The image size is 610 × 340 pixels with 102 spectral bands (430–680 nm). The spatial resolution is 1.3 m, and the spectral resolution is 3.74 nm. A total of nine land-cover classes are annotated.

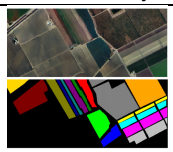
TABLE I
IPS, SV, PU, AND HS DATASETS

Indian Pines



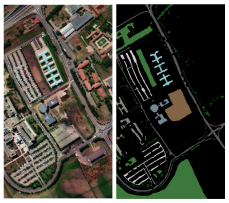
No.	Color	Class name	Samples	Proportion
C1		Alfalfa	46	0.45%
C2		Corn-no till	1428	13.93%
C3		Corn-min till	830	8.10%
C4		Corn	237	2.31%
C5		Grass/Pasture	483	4.71%
C6		Grass/Trees	730	7.12%
C7		Grass/pasture-mowed	28	0.27%
C8		Hay-windrowed	478	4.66%
C9		Oats	20	0.20%
C10		Soybeans-no till	972	9.48%
C11		Soybeans-min till	2455	23.95%
C12		Soybeans-clean till	593	5.79%
C13		Wheat	205	2.00%
C14		Woods	1265	12.34%
C15		Bldg-Grass-Tree-Drives	386	3.77%
C16		Stone-steel towers	93	0.91%
-		Background	10776	-

Salinas Valley



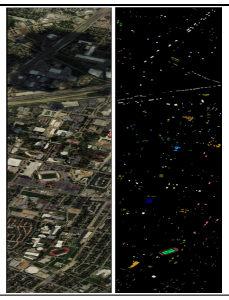
No.	Color	Class name	Samples	Proportion
C1		Brocoli_green_weeds 1	2009	3.71%
C2		Brocoli_green_weeds 2	3726	6.88%
C3		Fallow	1976	3.65%
C4		Fallow_rough_plow	1394	2.58%
C5		Fallow_smooth	2678	4.95%
C6		Stubble	3959	7.31%
C7		Celery	3579	6.61%
C8		Grapes_untrained	11271	20.82%
C9		Soil_vinyard_develop	6203	11.46%
C10		Corn_senesced_weeds	3278	6.06%
C11		Lettuce_romaine_4wk	1068	1.97%
C12		Lettuce_romaine_5wk	1927	3.56%
C13		Lettuce_romaine_6wk	916	1.69%
C14		Lettuce_romaine_7wk	1070	1.98%
C15		Vinyard_untrained	7268	13.43%
C16		Vinyard_vertical_trellis	1807	3.34%
-		Background	56975	-

Pavia University



No.	Color	Class name	Samples	Proportion
C1		Asphalt	6631	15.50%
C2		Meadows	18649	43.60%
C3		Painted metal sheets	2099	4.91%
C4		Gravel	3064	7.16%
C5		Trees	1345	3.14%
C6		Shadow	5029	11.76%
C7		Bare Soil	1330	3.11%
C8		Bitumen	3682	8.61%
C9		Bricks	947	2.21%
-		Background	164624	-

Houston



No.	Color	Class name	Samples	Proportion
C1		Healthy grass	1251	8.32%
C2		Stressed grass	1254	8.34%
C3		Synthetic grass	697	4.64%
C4		Trees	1244	8.28%
C5		Soil	1242	8.26%
C6		Water	325	2.16%
C7		Residential	1268	8.44%
C8		Commercial	1244	8.28%
C9		Road	1252	8.33%
C10		Highway	1227	8.16%
C11		Railway	1235	8.22%
C12		Parking Lot 1	1233	8.20%
C13		Parking Lot 2	469	3.12%
C14		Tennis Court	428	2.85%
C15		Running Track	660	4.39%
-		Background	649816	-

B. Experimental Settings

We compare the proposed method with several representative HSI classification models. For all four datasets, we report per-class accuracy, overall accuracy (OA), average accuracy (AA), and Kappa coefficient.

We train all models for a maximum of 200 epochs using the Adam optimizer. The initial learning rate is set to 0.005 and decays by a factor of 0.1 every 50 epochs. For the HS dataset, 50 labeled samples per class are used for training; for the remaining datasets, 100 labeled samples per class are randomly selected for training. When the total number of samples does not satisfy the above requirements, half of the total samples in that class are taken. Following common practice in HSI classification, the training and test samples are randomly selected from the entire image, which allows whole-image propagation models to fully exploit global contextual information Although random sampling may introduce spatial adjacency between training and test samples, it remains widely adopted to ensure consistent and fair comparison across different modeling paradigms, especially for whole-image propagation frameworks that inherently process the entire scene. Before training, PCA is applied to each dataset to retain 16 principal components.

To comprehensively evaluate the classification and generalization capabilities of the proposed FMGCN model, we compare it with several representative models, including SVM [8], classical convolutional-recurrent hybrid models (e.g., multiple vision architectures-based hybrid network (MVAHN) [48], full model (FM) [49]), attention multihop graph and multiscale convolutional fusion network (AMGCFN) [50], and dual-view spectral and global spatial feature fusion network (DSGSF) [51]. To assess the efficiency and accuracy of FMGCN, we further include some lightweight models such as channel-layer-oriented lightweight spectral–spatial network (CLOL) [52] and attention-assisted cascaded bottleneck network (ACB) [53], as well as recent deep learning architectures used in remote sensing tasks. Examples include mixed residual convolutions with ViT (MRViT) [54], Mamba [37], spectral–spatial global graph reasoning network (SSGRN) [19]. These models cover pixel-level, local-level, global-level modeling, and multiscale fusion paradigms, forming a representative technical lineage in the field of HSI classification. It is worth noting that Mamba and AMGCFN are consistent with the proposed model, both of which input the entire HSI in a single propagation.

All models are trained independently under the same configuration. The accuracy is computed by repeating each experiment five times with different random seeds to ensure reliability.

C. Results and Analysis

For the agricultural-scene datasets IP and SV, the classification results are shown in Figs. 4 and 5 and Tables II and III. FMGCN achieves the highest OA, demonstrating its better classification capability on agricultural hyperspectral scenes. The SVM classifier, which relies primarily on spectral

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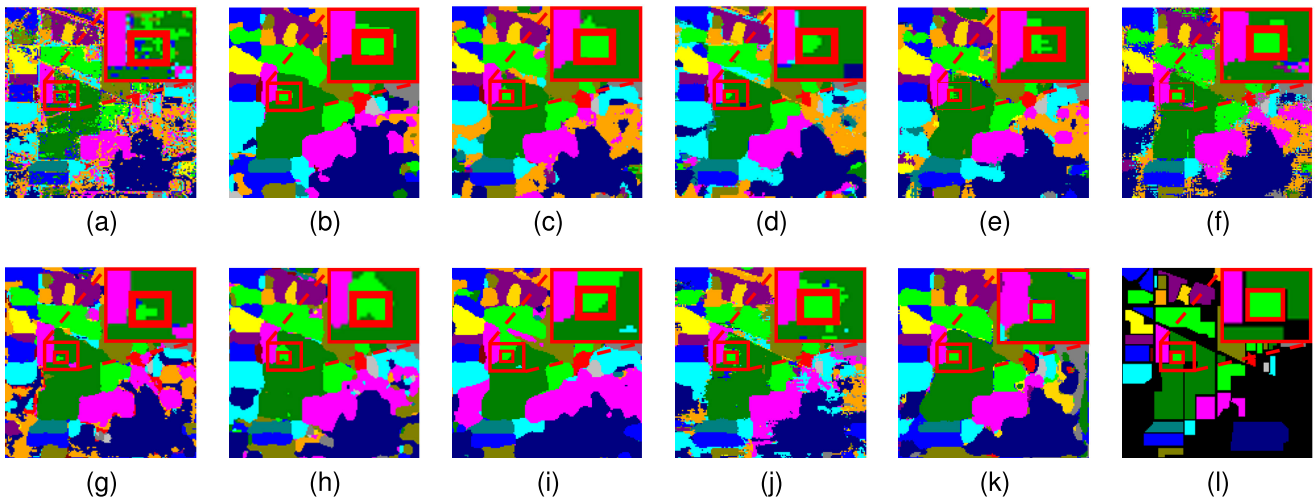


Fig. 4. Classification maps of different methods for the IP dataset. (a) SVM. (b) AMGCFN. (c) MVAHN. (d) DSGSF. (e) CLOL. (f) ACB. (g) MRViT. (h) Mamba. (i) SSGRN. (j) FM. (k) FMGCN. (l) Ground truth.

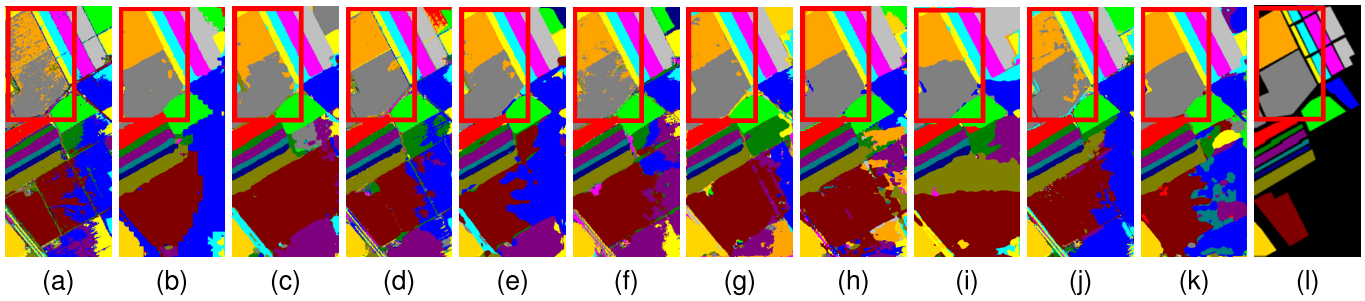


Fig. 5. Classification maps of different methods for the SV dataset. (a) SVM. (b) AMGCFN. (c) MVAHN. (d) DSGSF. (e) CLOL. (f) ACB. (g) MRViT. (h) Mamba. (i) SSGRN. (j) FM. (k) FMGCN. (l) Ground truth.

TABLE II

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE IP DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	AMGCFN	MVAHN	DSGSF	CLOL	ACB	MRViT	Mamba	SSGRN	FM	FMGCN
1	82.43%	**100.00**	89.37	**100.00**	**100.00**	**100.00**	99.13	96.63	**100.00**	99.82	99.16
2	76.69	96.64	98.69	98.65	88.86	97.71	96.47	95.93	86.14	96.61	98.11
3	76.51	97.49	94.20	98.41	98.49	**100.00**	95.92	99.73	98.63	97.51	**100.00**
4	88.74	94.43	93.14	81.25	**100.00**	98.32	92.85	**100.00**	**100.00**	99.98	**100.00**
5	93.38	98.79	98.12	98.59	96.61	99.59	99.30	95.25	96.08	**99.91**	98.81
6	91.19	98.05	93.18	96.30	**100.00**	97.69	99.31	**100.00**	98.25	99.88	**100.00**
7	86.11	**100.00**	98.67	86.67	**100.00**	**100.00**	96.47	**100.00**	**100.00**	99.84	**100.00**
8	97.06	99.61	99.55	**100.00**	**100.00**	99.58	99.55	**100.00**	**100.00**	98.59	**100.00**
9	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	87.47	**100.00**	**100.00**	99.94	**100.00**
10	81.24	91.26	90.83	90.45	90.25	98.12	89.40	97.89	97.48	98.80	**99.27**
11	68.87	99.25	98.83	**99.28**	92.02	95.74	99.11	98.56	85.52	95.48	99.00
12	79.80	98.90	98.01	95.85	93.91	97.31	97.15	98.98	97.97	98.44	**99.75**
13	97.68	98.33	99.19	99.01	**100.00**	**100.00**	99.73	**100.00**	**100.00**	**100.00**	**100.00**
14	92.93	99.76	99.68	**99.91**	99.14	99.70	99.69	99.56	98.88	99.56	99.75
15	71.48	94.26	94.37	95.29	99.14	97.93	91.38	**100.00**	**100.00**	96.51	**100.00**
16	93.00	84.62	92.96	84.91	**100.00**	95.74	96.54	96.06	**100.00**	98.38	99.02
OA (%)	80.56	97.47	96.87	97.32	94.56	97.62	97.01	98.43	93.20	97.08	**99.25**
AA (%)	86.07	96.96	96.17	95.29	97.41	98.59	96.22	98.66	92.16	98.70	**99.55**
Kappa	0.7803	97.09	0.9641	0.9689	0.9372	0.9712	0.9657	0.9818	0.9744	0.9666	**0.9912**

discrimination, performs significantly worse than most deep models that incorporate spatial information. Other deep learning models that integrate spectral and spatial information have all achieved relatively high classification accuracy, but they show clear differences in terms of consistency in classification maps, as well as differences in boundary discrimination. This is due to the varying degrees of dependence of different models on spatial and spectral information.

Models such as Mamba, DSGSF, and AMGCFN, which also emphasize global spatial modeling, achieve over 98%

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TABLE III
QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE SV DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	AMGCFN	MVAHN	DSGSF	CLOL	ACB	MRViT	Mamba	SSGRN	FM	FMGCN
1	99.63	**100.00**	**100.00**	99.95	99.63	**100.00**	99.97	**100.00**	99.36	99.27	**100.00**
2	99.53	**100.00**	99.99	99.83	99.99	99.94	99.96	**100.00**	95.53	**100.00**	**100.00**
3	99.63	**100.00**	99.75	97.86	99.95	**100.00**	**100.00**	**100.00**	98.15	99.89	**100.00**
4	99.77	99.75	98.91	97.12	99.54	99.57	99.88	99.92	99.45	**99.92**	99.85
5	96.51	99.57	**99.87**	99.55	99.63	99.76	99.41	98.75	97.86	99.65	99.22
6	99.53	99.99	**100.00**	99.87	**100.00**	99.96	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
7	99.37	99.95	99.99	99.68	99.60	99.91	**100.00**	99.97	98.53	99.37	99.83
8	81.18	96.15	96.59	**99.62**	94.08	91.88	96.22	96.30	78.52	94.37	99.45
9	97.51	**100.00**	99.75	99.60	99.89	99.92	99.98	**100.00**	99.41	99.77	**100.00**
10	93.61	99.18	98.88	94.68	98.79	98.63	99.18	99.65	96.44	**99.69**	99.02
11	95.35	**100.00**	99.93	96.06	99.74	**100.00**	98.57	**100.00**	98.83	99.48	**100.00**
12	99.89	**100.00**	99.97	99.66	99.92	99.97	99.89	**100.00**	99.39	99.84	**100.00**
13	98.28	**100.00**	99.94	99.23	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
14	94.54	99.35	99.97	96.75	99.90	**100.00**	99.09	**100.00**	99.87	99.79	**100.00**
15	69.00	99.39	90.04	95.04	90.58	94.08	99.53	99.25	97.40	86.02	**99.67**
16	99.41	**100.00**	**100.00**	97.66	99.41	50.32	**100.00**	99.88	**100.00**	99.18	**100.00**
OA (%)	90.54	99.01	97.72	98.39	97.27	97.26	98.96	99.70	97.97	96.71	**99.73**
AA (%)	95.17	99.58	98.97	98.26	98.79	98.96	99.34	99.81	97.72	98.52	**99.82**
Kappa	0.8945	98.90	0.9746	0.9820	0.9695	0.9694	0.9884	0.9967	0.9862	0.9633	**0.9969**

accuracy, highlighting the importance of global information in these datasets. However, AMGCFN, Mamba, and SSGRN, which process the entire image in a single pass, exhibit severe over-smoothing effects, most likely caused by the over-domination of global spatial information that suppresses local spectral variability. FMGCN is also a whole-image propagation model, yet its classification maps show sharper bounds compared to AMGCFN while retaining the advantage of global inference. This improvement highlights the effectiveness of both the proposed SCRN and the EECS in recovering high-frequency information. Nonetheless, these results also reveal a shared limitation across global propagation models: distinguishing unlabeled background pixels without explicit supervision remains challenging and warrants further investigation.

Thanks to their strong local context modeling capabilities, MRViT and FM also achieve high classification accuracy, although their reliance on long-range attention mechanisms results in increased computational cost. CLOB and ACB achieve competitive accuracy with low computational overhead, but their mini-batch inference design limits the actual inference time as they require sequential batching.

Agricultural scene classification typically emphasizes regional segmentation. Due to the high spectral resolution and coarse spatial resolution of remote sensing HSIs, each pixel often contains mixed spectral signatures of multiple land-cover types. In agricultural environments, factors such as soil, vegetation, and moisture variability create significant intraclass spectral variability. Therefore, annotations often follow region-based labeling, where pixels within a crop region are assigned the same class. Consequently, models that emphasize spatial aggregation tend to outperform models that focus primarily on spectral discrimination. This explains why SVMs are prone to salt-and-pepper noise and have lower accuracy, while models that focus on spatial features perform better on both IP and SV datasets.

The results for the urban datasets (PU and HS) are shown in Figs. 6 and 7 and Tables IV and V. Unlike agricultural scenes, urban environments exhibit richer high-frequency structures and more complex object boundaries. Spectral-focused models achieve sharper boundary delineation, whereas spatial-focused models are more prone to over-smoothing. In addition, the clustered nature of urban objects also increases the likelihood of salt-and-pepper noise when relying too heavily on spectral cues.

AMGCFN suffers from a significant over-smoothing effect. Its whole-image propagation deepens the model's reliance on spatial features and suppresses spectral sharpness. This leads to diffusive-like label propagation and degenerate boundary classification. Similarly, as a single whole-image propagation model, SSGRN also shows significant over-smoothing. FMGCN also shows a smoothing tendency (typical of whole-image propagation models), but significantly improves the quality of the bounds compared to AMGCFN and SSGRN. This again validates the benefits of the proposed EECS and SCRN.

The 2-D and 3-D convolution-based models jointly extract spectral–spatial information, and their visualization maps reflect a different balance between spectral and spatial information. For example, FM, SVM, and MRViT preserve richer spectral variability, leading to sharper bounds. However, batch-wise training prevents these models from fully exploiting global spatial relationships, limiting their ultimate accuracy. Moreover, batching requires repeated loading and propagation during testing, resulting in significantly longer inference times. FMGCN avoids these limitations by performing single-pass whole-image propagation. By integrating EECS and SCRN, FMGCN mitigates over-smoothing and achieves the best accuracy with fast whole-image inference.

To further evaluate robustness, we varied the number of labeled samples per class to 25, 50, 75, 100, and 150. All other settings remain the same. The results across the four datasets

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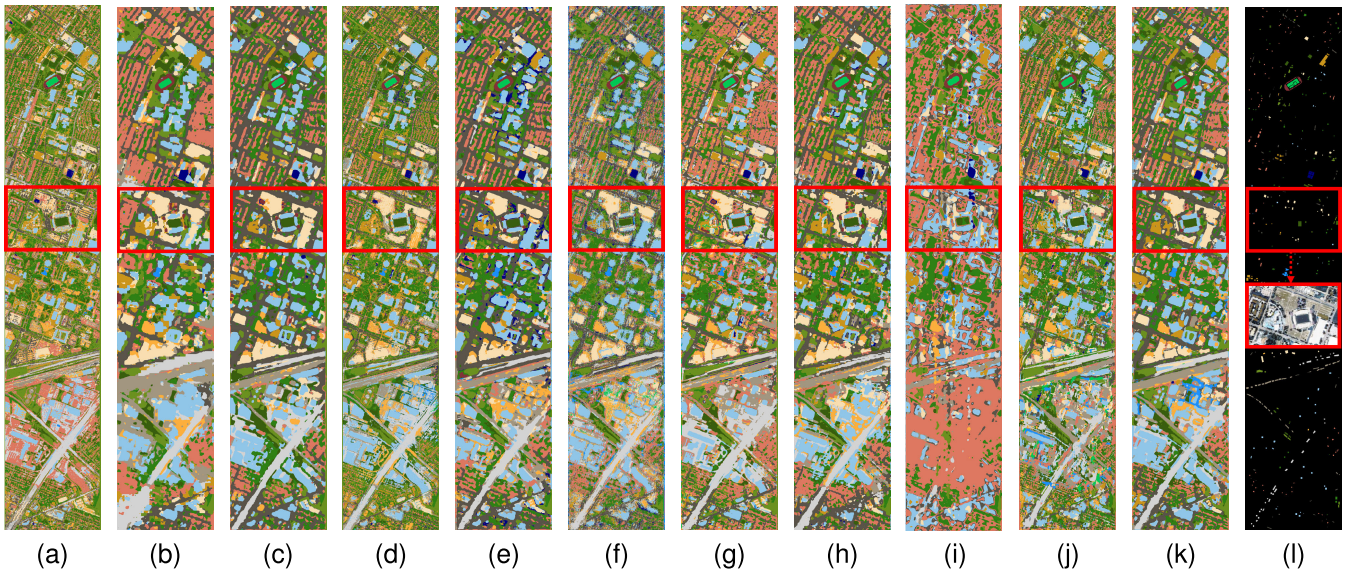


Fig. 6. Classification maps of different methods for the HS dataset. (a) SVM. (b) AMGCFN. (c) MVAHN. (d) DSGSF. (e) CLOL. (f) ACB. (g) MRViT. (h) Mamba. (i) SSGRN. (j) FM. (k) FMGCN. (l) Ground truth.

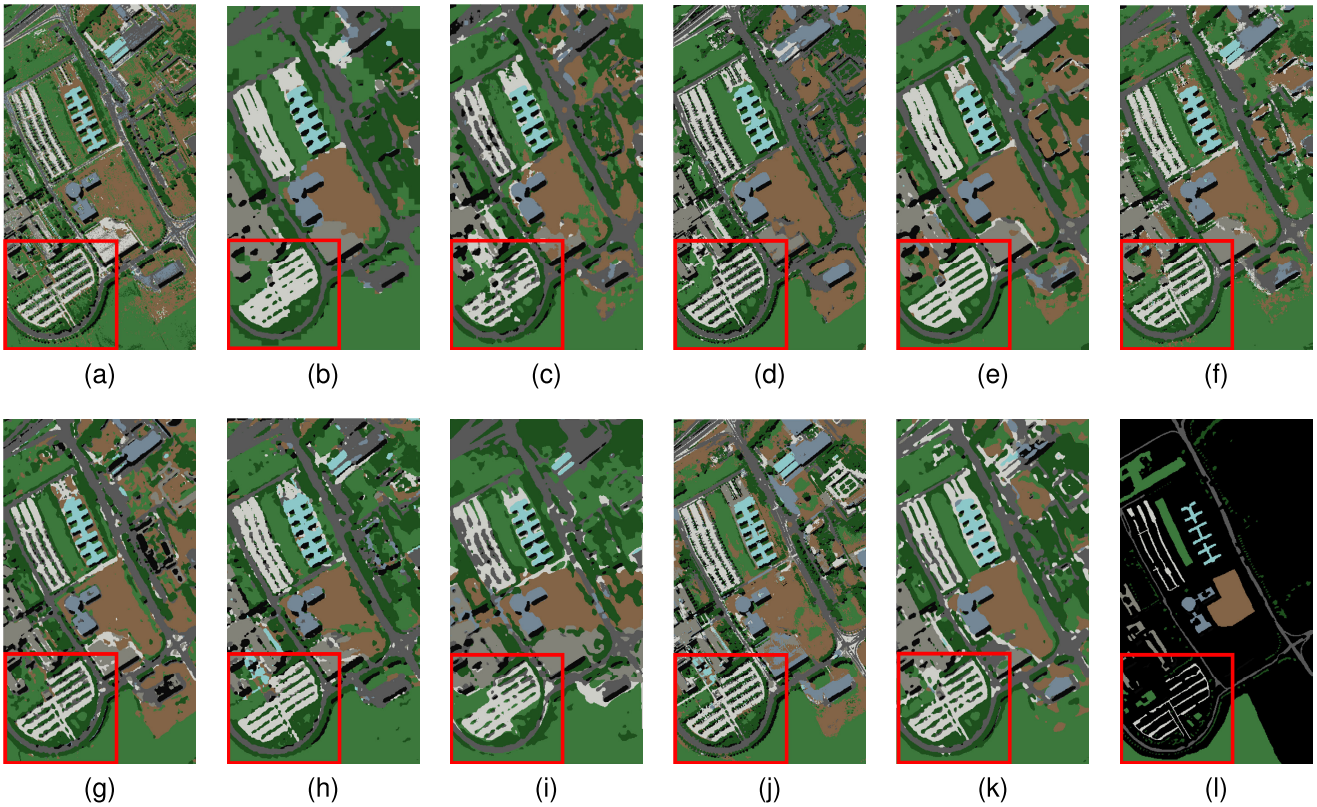


Fig. 7. Classification maps of different methods for the PU dataset. (a) SVM. (b) AMGCFN. (c) MVAHN. (d) DSGSF. (e) CLOL. (f) ACB. (g) MRViT. (h) Mamba. (i) SSGRN. (j) FM. (k) FMGCN. (l) Ground truth.

are shown in Fig. 8. FMGCN achieves the best accuracy for most training sample sizes, confirming its better classification and generalization ability.

When the number of labeled samples is extremely small, the performance degrades compared to several baselines. This is most likely due to insufficient supervision, which makes global modeling more difficult and reduces the effectiveness of edge compensation. Improving the performance in few-shot scenarios will be an important direction for future research.

D. Key Parameter Experiments

MKGC is the core operator in FMGCN. It contains two key parameters: the number of kernels and the feature dimension of the nodes. These parameters determine the number of output

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TABLE IV

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE HS DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	AMGCFN	MVAHN	DSGSF	CLOL	ACB	MRViT	Mamba	SSGRN	FM	FMGCN
1	93.84	96.61	**99.37**	97.33	92.09	92.77	98.48	94.71	85.58	89.18	98.98
2	98.75	98.63	**99.91**	98.96	99.74	94.21	99.41	97.24	94.02	88.87	99.49
3	99.38	99.90	**100.00**	99.56	99.33	99.60	**100.00**	99.84	99.50	**100.00**	**100.00**
4	92.46	98.98	96.92	98.54	97.38	94.25	**99.83**	96.03	99.04	95.90	99.74
5	97.65	99.93	95.89	97.85	99.91	98.94	**100.00**	99.83	99.21	93.71	99.65
6	97.82	99.14	99.24	98.12	**99.56**	96.00	98.71	99.25	95.56	96.00	99.13
7	90.07	96.05	**99.02**	95.74	89.90	91.76	93.43	96.36	84.50	93.35	96.12
8	54.19	91.29	**100.00**	97.59	93.09	91.95	94.67	87.08	86.54	78.64	95.94
9	76.62	94.23	96.59	**96.73**	95.05	90.68	96.05	90.44	84.64	86.19	96.29
10	78.93	99.22	99.12	98.08	96.98	98.73	99.66	**99.91**	84.56	93.63	99.83
11	63.46	98.53	99.01	98.19	97.00	95.17	98.38	92.60	91.01	94.35	**99.10**
12	54.35	97.57	93.89	97.94	95.68	96.42	98.12	95.91	91.17	86.81	**99.06**
13	49.40	98.06	91.18	95.76	95.39	94.80	97.76	94.38	96.21	**99.50**	98.74
14	97.09	99.88	97.57	98.23	**100.00**	**100.00**	99.75	**100.00**	98.48	**100.00**	**100.00**
15	98.52	**100.00**	96.35	99.22	**100.00**	**100.00**	**100.00**	**98.83**	98.93	99.84	**100.00**
OA (%)	81.66	97.49	97.73	97.79	96.17	95.02	98.07	95.58	91.27	91.58	**98.62**
AA (%)	82.84	97.87	97.60	97.86	96.74	95.69	98.28	96.16	92.60	93.07	**98.80**
Kappa	0.8016	97.29	0.9755	0.9761	0.9585	0.9459	0.9791	0.9521	0.9055	0.9089	**0.9851**

TABLE V

QUANTITATIVE PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS IN TERMS OF OA, AA, AND KAPPA ON THE PU DATASET. THE BEST ONE IS SHOWN IN BOLD

Class	SVM	AMGCFN	MVAHN	DSGSF	CLOL	ACB	MRViT	Mamba	SSGRN	FM	FMGCN
1	82.59	97.93	98.68	**99.09**	97.18	98.17	98.81	98.48	89.55	97.21	98.75
2	89.82	98.35	**99.82**	99.59	98.57	94.74	99.78	98.66	98.12	96.57	99.73
3	84.04	99.56	96.61	94.42	**99.90**	98.67	96.05	99.30	95.16	96.85	99.06
4	96.32	97.26	96.85	98.83	97.91	99.10	97.65	97.36	96.35	**99.22**	99.03
5	99.36	**100.00**	99.51	97.72	**100.00**	**100.00**	99.87	**100.00**	99.90	99.76	**100.00**
6	88.25	**100.00**	99.34	96.78	99.80	98.98	98.96	**100.00**	**100.00**	97.04	**100.00**
7	95.04	**100.00**	95.23	96.89	**100.00**	99.90	96.53	**100.00**	**100.00**	99.59	99.90
8	86.54	**99.70**	92.85	94.99	98.77	98.11	93.63	98.57	96.39	92.55	98.83
9	99.88	97.61	96.24	99.02	98.94	99.54	98.01	**100.00**	99.54	**100.00**	**100.00**
OA (%)	89.05	98.66	98.40	98.32	98.62	96.89	98.53	98.82	96.71	96.83	**99.46**
AA (%)	91.32	98.93	97.24	97.48	99.01	98.58	97.70	99.15	97.22	97.64	**99.47**
Kappa	0.8563	98.22	0.9787	0.9776	0.9817	0.9582	0.9804	0.9843	0.9555	0.9579	**0.9928**

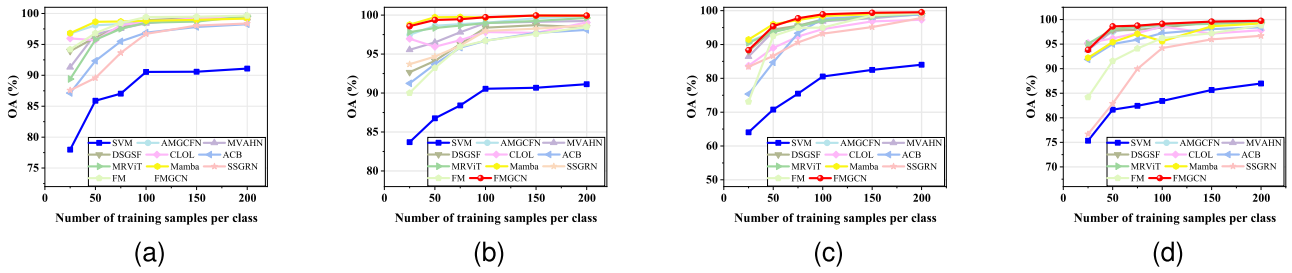


Fig. 8. Classification accuracy of different methods under different scales of training samples. (a) PU. (b) SV. (c) IP. (d) HS.

channels and the size of the feature maps. To investigate their contributions, we evaluate the first two MKGC layers using different kernel numbers and feature dimensions on the SV dataset, with 50 labeled samples per class and all other settings unchanged. The parameters of the third MKGC are constrained by the input size of the subsequent decoder. Therefore, it was not included in this experiment.

The results are shown in Fig. 9. Both parameters exhibit a nonmonotonic relationship with performance: too few kernels weaken the representation capacity, whereas too many increase the model complexity and optimization difficulty. The feature dimensions of the nodes show similar characteristics. Optimal performance is achieved when both parameters are in the range 4–6, balancing network performance with computational tractability.

E. Ablation and Efficiency Experiments

We further conduct a quantitative analysis of computational efficiency to substantiate the claimed advantage of FMGCN in whole-image hyperspectral classification. We evaluate the storage cost (number of parameters), computational

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TABLE VI

COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS, WITH THE BEST RESULT HIGHLIGHTED IN BOLD AND THE SECOND BEST UNDERLINED

	FLOPS(G)				Params (M)				Times (s)
Dataset	PU	SV	IP	HS	PU	SV	IP	HS	PU
AMGCFN	14.98	10.92	2.05	56.66	0.32	0.34	0.34	0.33	**0.042**
MVAHN	5.25	7.93	7.83	27.56	0.25	0.38	0.37	0.36	20.52
DSGSF	8.46	4.60	0.89	27.16	0.39	0.40	0.40	0.40	44.86
ACB	**0.10**	**0.10**	**0.10**	**0.10**	0.018	0.022	0.022	0.021	2.91
CLOL	0.16	0.23	0.22	0.19	**0.0035**	**0.0053**	**0.0052**	**0.0043**	7.36
MRVIT	684.86	717.72	717.72	717.72	27.06	28.07	28.07	28.07	27.06
Mamba	20.04	12.17	2.29	69.65	0.18	0.19	0.19	0.18	0.93
SSGRN	101.55	58.89	10.70	396.12	32.39	10.84	2.19	316.05	0.97
FM	160.30	103.25	103.22	102.86	14.50	17.83	17.70	15.85	402.99
FMGCN	51.73	33.59	1.44	55.44	0.37	0.37	0.37	0.38	0.44

TABLE VII

ABLATION EXPERIMENT OF FMGCN. THE BEST ONE IS SHOWN IN BOLD

	REMOVE	MKGCN	EDGE	CSRN	SR	MKGC	SLIC	Conv$_{adj}$	FMGCN
UP	OA (%)	95.92	96.34	95.13	96.06	96.27	95.31	96.17	**96.81**
	AA (%)	96.50	97.17	94.56	96.66	96.78	95.47	96.43	**97.24**
	Kappa	0.9525	0.9560	0.9357	0.9479	0.9612	0.9482	0.9579	**0.9688**
SV	OA (%)	98.60	98.86	98.74	99.07	98.88	98.72	99.03	**99.38**
	AA (%)	98.55	99.28	98.88	99.13	99.01	99.13	98.74	**99.43**
	Kappa	0.9855	0.9872	0.9868	0.9899	0.9872	0.9849	0.9864	**0.9930**
IP	OA (%)	95.17	94.52	94.22	94.63	94.89	95.22	94.57	**95.41**
	AA (%)	96.02	95.22	96.31	97.05	96.27	96.84	96.62	**97.25**
	Kappa	0.9359	0.9273	0.9228	0.9312	0.9407	**0.9517**	0.9443	0.9472
HS	OA (%)	98.01	97.95	98.15	98.22	98.44	97.62	98.22	**98.62**
	AA (%)	98.52	98.29	98.23	98.75	98.38	98.12	98.47	**98.80**
	Kappa	0.9772	0.9728	0.9757	0.9763	0.9816	0.9706	0.9793	**0.9851**

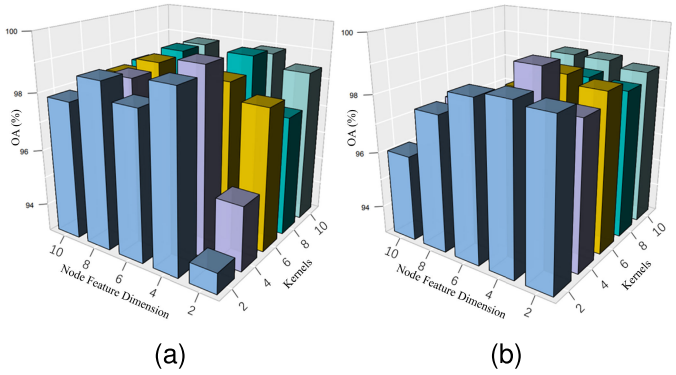


Fig. 9. Classification accuracy under different key parameter combinations. (a) First MKGC layer. (b) Second MKGC layer.

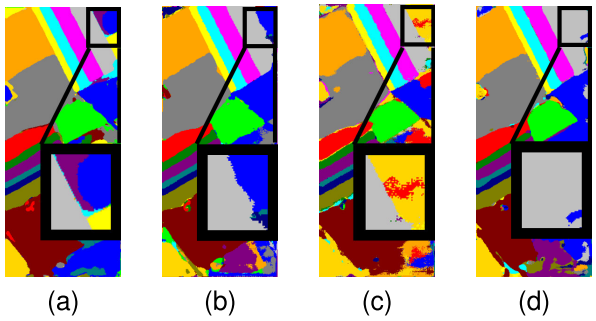


Fig. 10. Visual ablation experiments of SCRN and EECS. (a) FMGCN. (b) w/o SCRN. (c) w/o EECS. (d) w/o SCRN and EECS.

cost (FLOPs), and whole-image inference time. Results are summarized in Table VI. Lightweight models such as ACB and CLOB have small parameter counts and low FLOPs, but their mini-batch inference strategy results in significantly longer inference time due to repeated patch extraction and redundant forward propagation. AMGCFN and Mamba achieve favorable inference time owing to their whole-image propagation manner, but both methods exhibit limited boundary preservation ability, as discussed in the qualitative analysis. Due to their complex hybrid structures, MRViT and FM incur extremely high computational costs and long inference times. MVAHN and DSGSF achieve moderate FLOPs but still inherit the efficiency bottleneck of batch-wise inference. As shown in Table VI, FMGCN consistently achieves lower whole-image inference time than batch-wise methods while maintaining moderate FLOPs and parameter counts across all evaluated datasets. From a complexity perspective, the proposed MKGC operates on a compressed superpixel-level graph, reducing the graph size from $O(\mathrm{HW})$ to $O(N_s)$, where $N_s \ll \mathrm{HW}$, while multikernel propagation is performed in parallel. This design effectively controls the overall computational complexity of whole-image graph convolution. More importantly, by performing single-pass whole-image inference on a compressed graph representation, FMGCN significantly reduces

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redundant computation and enables fast end-to-end prediction. This quantitative comparison demonstrates that FMGCN provides a favorable tradeoff among accuracy, efficiency, and structural fidelity, making it suitable for large-scale real-world HSI applications.

To validate the effectiveness of each module, we conduct ablation experiments on all four datasets using 50 samples per class. All other hyperparameters follow the default settings. As shown in Table VII, removing MKGCN leads to a significant drop in accuracy, which validates that it can extract deeper features more fully and effectively improve classification performance. Removing SCRN or EECS also results in performance degradation, highlighting their importance in recovering high-frequency information. In ablation settings, replacing SRM or SLIC with pooling leads to reduced accuracy. SRM retains more deep-level spectral information, which is particularly valuable for reconstruction networks where the input already contains rich and valid information. SLIC can extract certain regional relationship information. Pooling, by contrast, inevitably removes spatial details and impairs classification. It is worth noting that the model achieves the best kappa on the IP dataset when pooling is used instead of SLIC. This phenomenon can be attributed to the fact that IP datasets inherently have more regular feature distributions. This distribution is more compatible with the square pooling operation, leading to competitive classification results compared to SLIC. The definition in (3) is consistent with Conv_{adj}. After removing this convolution, the accuracy decreases slightly. This may be due to the fact that the adjacency matrix cannot be updated during construction, making it difficult to adapt more flexibly to the modeling of the relationships of multiple kernels. Furthermore, we replace MKGC with a regular single-kernel graph convolution process to more intuitively explore the role of multiple kernels in graph convolution. Experimental results show that accuracy decreases on multiple datasets when single-kernel graph convolution is used. To a certain extent, it has verified the effectiveness of its more comprehensive relationship modeling in fully exploring the features. Moreover, Fig. 10 visualizes the results on the SV dataset and compares the two compensation methods. The compensated outputs show improved boundary preservation, thus validating the effectiveness of both algorithms.

IV. Conclusion

In this article, we propose FMGCN, a framework for fast whole-image classification of HSIs. By integrating the deep spectral modeling capability of MKGC, the structural information reinforcement mechanism of SCRN, and the high-frequency detail enhancement mechanism of EECS, we achieve efficient and high-precision whole-image inference within a unified framework. Thanks to the advantage of MKGC in modeling spectral dependencies in non-Euclidean space, the network is able to obtain richer representations of contextual relations in a single propagation. At the same time, SCRN and EECS further compensate for the inherent imperfections of the weakened high-frequency components in the propagation of the whole image, enabling the model to recover local detail information more accurately while maintaining global consistency. Relying on this family of designs, FMGCN achieves fast whole-image classification, effectively reducing inference overhead while preserving competitive classification performance.

On the four representative datasets, IP, SV, HS, and UP, FMGCN achieves the best classification performance. Experimental results show that FMGCN reduces over-smoothing and is more stable and reliable in expressing boundary features in scenarios with complex spatial structures or rich high-frequency boundary information compared to typical whole-image propagation models. Compared to batch-wise inference models, it demonstrates stronger global dependency modeling capabilities and significantly shorter inference times. Comparative experiments with different numbers of training samples also show that FMGCN maintains better classification and generalization performance and can adapt to different data scales and scenario requirements.

Despite the notable performance improvement achieved by the proposed model, there is still room for further optimization of efficient mining of global relationships under minimal sample conditions, and category discrimination in unlabeled backgrounds remains a common challenge for whole-image propagation methods. In addition, extending the proposed framework to large-scale hyperspectral scenes with higher spatial resolution and larger graph sizes poses new challenges in terms of scalability and memory efficiency. Another important direction is to improve the robustness of the model under spectral variability caused by illumination variations, atmospheric effects, or sensor discrepancies, which is particularly critical for cross-scene and long-term Earth observation applications. From an evaluation perspective, the impact of different data partition strategies on whole-image propagation models deserves further investigation, especially regarding information leakage and fairness in model comparison under spatially disjoint settings. From a methodological perspective, future work may explore more adaptive graph construction and learning strategies, such as dynamic or learnable adjacency modeling, to further enhance the flexibility of spectral–spatial relationship modeling. Moreover, incorporating recent state-space-based architectures into graph learning frameworks could provide a promising avenue to capture long-range dependencies while maintaining computational efficiency. Future research will therefore focus on more efficient high-frequency compensation mechanisms, weakly supervised or self-supervised learning strategies, and graph structure optimization suitable for large-scale remote sensing scenarios to further enhance the applicability, stability, and classification accuracy of FMGCN in complex real-world scenarios. while also investigating the impact of data partition strategies and potential information leakage on the evaluation of whole-image propagation models.

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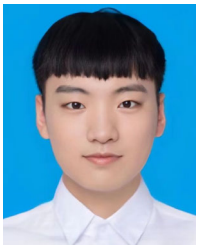
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